

Vectors: Components and Operations

ANSWER KEY



Vector arithmetic

- 1 Let $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 4 \rangle$. Find:

a $\mathbf{u} + \mathbf{v} = \langle 2, 2 \rangle$

b $3\mathbf{u} - 2\mathbf{v} = \langle 11, -14 \rangle$

c $|\mathbf{u}| = \sqrt{13}$

- 2 Find the component form and magnitude of the vector from $P(1, 3)$ to $Q(5, -2)$.

$\vec{PQ} = \langle 4, -5 \rangle$, $|\vec{PQ}| = \sqrt{16 + 25} = \sqrt{41}$.

- 3 Find the unit vector in the direction of $\mathbf{v} = \langle 6, -8 \rangle$.

$|\mathbf{v}| = 10$, so $\hat{\mathbf{v}} = \langle \frac{3}{5}, -\frac{4}{5} \rangle$.

- 4 Write the vector of magnitude 12 making an angle of 150° with the positive x-axis in component form (exact values).

$\langle 12\cos 150^\circ, 12\sin 150^\circ \rangle = \langle -6\sqrt{3}, 6 \rangle$.

Applications

- 5 A plane flies due east at 200 mph while the wind blows due north at 50 mph.

a Find the magnitude of the plane's resultant velocity, to one decimal place.

$\sqrt{200^2 + 50^2} = \sqrt{42500} \approx 206.2$ mph.

b Find the direction of the resultant, as an angle north of east to two decimal places.

$\theta = \tan^{-1}\left(\frac{50}{200}\right) \approx 14.04^\circ$ north of east.

- 6 Two forces $\mathbf{F}_1 = \langle 30, 0 \rangle$ N and $\mathbf{F}_2 = \langle 0, 40 \rangle$ N act on an object. Find the magnitude of the resultant force.

$|\mathbf{F}_1 + \mathbf{F}_2| = |\langle 30, 40 \rangle| = 50$ N.

- 7 A boat aims to cross a river by heading due north at 4 m/s, but the current pushes it east at 1.5 m/s.

a Find the boat's resultant speed, to two decimal places. $\sqrt{4^2 + 1.5^2} = \sqrt{18.25} \approx 4.27$ m/s.

b If the river is 200 m wide, find how long the crossing takes. Only the northward component moves it across: $t = \frac{200}{4} = 50$ s.

c Find how far downstream (east) the boat lands. Drift = $1.5 \times 50 = 75$ m.

Position vectors and geometry

- 8 Given $A(-2, 5)$ and $B(4, -3)$, find the midpoint of $\overline{AB}$ and the vector $\vec{AB}$.

Midpoint: $\left(\frac{-2+4}{2}, \frac{5-3}{2}\right) = (1, 1)$. $\vec{AB} = \langle 6, -8 \rangle$.

- 9 Let $\mathbf{a} = \langle 2, 3 \rangle$ and $\mathbf{b} = \langle -4, 1 \rangle$. Find a vector $\mathbf{c}$ such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \langle 0, 0 \rangle$, and verify your answer.
 $\mathbf{c} = -(\mathbf{a} + \mathbf{b}) = -\langle -2, 4 \rangle = \langle 2, -4 \rangle$. Check: $\langle 2, 3 \rangle + \langle -4, 1 \rangle + \langle 2, -4 \rangle = \langle 0, 0 \rangle$. ✓
- 10 Express $\mathbf{v} = \langle 5, -12 \rangle$ in terms of the standard unit vectors $\mathbf{i}$ and $\mathbf{j}$, and find $|\mathbf{v}|$.
 $\mathbf{v} = 5\mathbf{i} - 12\mathbf{j}$. $|\mathbf{v}| = \sqrt{25 + 144} = 13$.

More applications

- 11 An airplane has an airspeed of 305 mph and flies on a heading 7.5° west of due north, while a 40 mph wind blows due east. Find the plane's resultant ground velocity vector, and its ground speed, to one decimal place (take east as the positive x-direction and north as the positive y-direction).
 Plane's own velocity: $\langle -305\sin 7.5^\circ, 305\cos 7.5^\circ \rangle \approx \langle -39.8, 302.4 \rangle$. Adding the wind $\langle 40, 0 \rangle$:
 resultant $\approx \langle 0.2, 302.4 \rangle$. Ground speed $\approx \sqrt{0.2^2 + 302.4^2} \approx 302.4$ mph — very nearly due north, as intended.
- 12 Given $\mathbf{u} = \langle 4, -1 \rangle$, find a vector orthogonal to $\mathbf{u}$ with the same magnitude as $\mathbf{u}$.
 Swap components and negate one: $\langle 1, 4 \rangle$. Check: $\mathbf{u} \cdot \langle 1, 4 \rangle = 4 - 4 = 0$ (orthogonal), and
 $|\langle 1, 4 \rangle| = \sqrt{17} = |\mathbf{u}|$.

Finding direction angles

- 13 Find the direction angle of $\mathbf{v} = \langle -3, 3\sqrt{3} \rangle$, to the nearest degree.
 $\theta = \tan^{-1}\left(\frac{3\sqrt{3}}{-3}\right)$ with reference angle 60° , but since the vector has negative x and positive y (Quadrant II): $\theta = 180^\circ - 60^\circ = 120^\circ$.
- 14 A vector has magnitude 10 and direction angle 200° . Write it in component form, rounded to two decimal places.
 $\langle 10\cos 200^\circ, 10\sin 200^\circ \rangle \approx \langle -9.40, -3.42 \rangle$.

Vector equations and unknowns

- 15 Find scalars c_1 and c_2 such that $c_1\langle 1, 0 \rangle + c_2\langle 0, 1 \rangle = \langle -3, 7 \rangle$.
 Since $\langle 1, 0 \rangle$ and $\langle 0, 1 \rangle$ are the standard basis vectors, $c_1 = -3$ and $c_2 = 7$ directly.
- 16 Find scalars c_1 and c_2 such that $c_1\langle 2, 1 \rangle + c_2\langle 1, 3 \rangle = \langle 7, 11 \rangle$.
 System: $2c_1 + c_2 = 7$ and $c_1 + 3c_2 = 11$. From the first: $c_2 = 7 - 2c_1$. Substituting:
 $c_1 + 3(7 - 2c_1) = 11 \Rightarrow c_1 + 21 - 6c_1 = 11 \Rightarrow -5c_1 = -10 \Rightarrow c_1 = 2$. Then $c_2 = 7 - 4 = 3$.

Parallel and equal vectors

- 17 Determine whether $\mathbf{u} = \langle 4, -6 \rangle$ and $\mathbf{v} = \langle -6, 9 \rangle$ are parallel, and if so, state whether they point in the same or opposite directions.
 $\mathbf{v} = -\frac{3}{2}\mathbf{u}$ (check: $-\frac{3}{2}(4) = -6$ ✓, $-\frac{3}{2}(-6) = 9$ ✓). Parallel, with a negative scalar multiple, so they point in OPPOSITE directions.
- 18 Find the value of k so that $\langle k, 6 \rangle$ is parallel to $\langle 4, 8 \rangle$.
 Need $\frac{k}{4} = \frac{6}{8} = \frac{3}{4}$: $k = 3$.

Vectors in three real-world scenarios

- 19 A hiker walks 3 km east, then 4 km north, then 2 km west. Find the magnitude of the hiker's total displacement from the starting point.
 Net displacement: $\langle 3 - 2, 4 \rangle = \langle 1, 4 \rangle$. Magnitude $= \sqrt{1 + 16} = \sqrt{17} \approx 4.12$ km.
- 20 Three ropes pull on a ring with forces $\langle 10, 0 \rangle$, $\langle -5, 8.66 \rangle$, and $\langle -5, -8.66 \rangle$ N. Find the net force, and explain what this implies physically.
 Sum: $\langle 10 - 5 - 5, 0 + 8.66 - 8.66 \rangle = \langle 0, 0 \rangle$. The net force is zero, meaning the ring is in equilibrium (the three forces are 120° apart and balance exactly).
- 21 A crane hoists a load using two cables. One cable exerts a force of $\langle -200, 350 \rangle$ N and the other exerts $\langle 200, 400 \rangle$ N. Find the combined lifting force (the vertical component of the resultant), and explain why the horizontal components matter for the crane's structural design even though they cancel.
 Resultant: $\langle -200 + 200, 350 + 400 \rangle = \langle 0, 750 \rangle$. The vertical (lifting) force is 750 N. Although the horizontal components cancel in the resultant, each individual cable still experiences its own horizontal tension, which the crane's frame must be strong enough to withstand even though the net horizontal effect on the load is zero.
- 22 A swimmer aims to swim directly across a river at 1.5 m/s relative to the water, while the current flows at 0.8 m/s downstream. Find the swimmer's resultant speed relative to the shore, and the angle (from the intended straight-across direction) at which they actually drift, to the nearest degree.
 Resultant speed $= \sqrt{1.5^2 + 0.8^2} = \sqrt{2.25 + 0.64} = \sqrt{2.89} = 1.7$ m/s. Drift angle $= \tan^{-1}\left(\frac{0.8}{1.5}\right) \approx 28^\circ$ from the intended direction.